

Design Justification Report

RC-DFA Memcapacitive Reservoir Accelerator

ECE 410/510 Hardware for Artificial Intelligence and Machine Learning — Spring 2026

Mohammad Alshaiji

Milestone 4 | June 7, 2026

1. Problem and Motivation

The target application is satellite image compression using a convolutional autoencoder trained via Direct Feedback Alignment with a Spatial Tiled Reservoir (RC-DFA). The encoder receives 4-channel RGBD patches of 128×128 pixels and compresses them to a 16×8×8 latent space, achieving a 64:1 compression ratio. Training uses a fixed random echo-state reservoir to project a global reconstruction error signal back to local weight gradients at each encoder layer, bypassing backpropagation's weight transport requirement. Figure 2 illustrates the DFA architecture.

CPU profiling on an Intel Core Ultra 9 275HX (cProfile, 3 batches, batch size 64) identified **SpatialTiledReservoir.get_feedback_map** as the dominant kernel, consuming **76.3% of total training step time** (6.091 s of 11.095 s total). The kernel performs $T=20$ recurrent echo-state updates over all spatial locations of each encoder layer's error map. Measured CPU throughput was 651 GFLOP/s at 47% of the theoretical FP32 peak (1,382 GFLOP/s AVX2). End-to-end training rate was 18.8 samples/sec (85.12 s/epoch, 25 steps/epoch, batch 64). These numbers are from `project/m1/sw_baseline.md` and reproduced by running `tiled_profile.py`.

Custom hardware acceleration is justified on three grounds. First, the kernel fraction (76.3%) is high enough that Amdahl's law bounds the maximum achievable speedup at $1 / (1 - 0.763) = 4.22\times$ — hardware must completely offload the kernel to approach this ceiling. Second, the reservoir weights W_{res} and W_{lat} are fixed at fabrication as device capacitances, eliminating all weight-loading memory traffic during inference. Third, the physical substrate — a memcapacitive RC crossbar — natively implements the continuous recurrent dynamics the algorithm requires, providing the echo-state behavior at the speed of charge redistribution rather than digital multiply-accumulate.

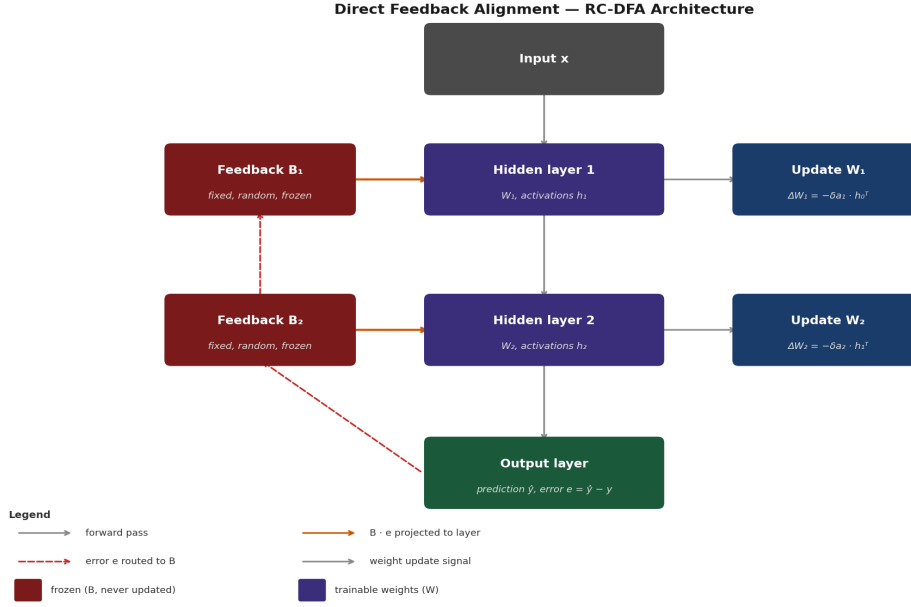


Figure 2: Direct Feedback Alignment (DFA) architecture. Fixed random feedback matrices B_1 , B_2 project the output error e directly to each hidden layer, replacing backpropagation. The reservoir implements the B matrices in analog hardware.

2. Roofline Analysis

The arithmetic intensity of the target kernel was computed from first principles. For one reservoir call over one spatial location with $N_{IN}=64$ input channels, $N_{RES}=128$ reservoir nodes, and $T=20$ recurrent steps: FLOPs = $2 \times N_{IN} \times N_{RES} \times T = 2 \times 64 \times 128 \times 20 = 327,680$ FLOP/call. Bytes transferred = $N_{IN} \times 4 = 64 \times 4 = 256$ bytes/call (M3 interface). Arithmetic intensity (M3) = $327,680 / 256 = 1,280$ FLOP/byte.

With the M4 interface ($N_{IN}=4,096$ channels per call): FLOPs = $2 \times 4096 \times 128 \times 20 = 20,971,520$ FLOP/call. Bytes = $4096 \times 4 = 16,384$ bytes/call. AI = $20,971,520 / 16,384 = \mathbf{1,280 \text{ FLOP/byte}}$ — identical to M3, because both compute and data scale linearly with N_{IN} .

The CPU ridge point is 15.4 FLOP/byte ($1,382 / 89.6$), confirming the kernel is compute-bound on the CPU. The hardware ridge point is 5,120 FLOP/byte ($32,768 \text{ GFLOP/s analog ceiling} / 6.4 \text{ GB/s UCle bandwidth}$). At AI=1,280 FLOP/byte, M4 sits on the memory-bound slope of the hardware roofline — the binding constraint is the UCle link, not the analog compute ceiling. The predicted achievable performance is $6.4 \text{ GB/s} \times 1,280 \text{ FLOP/byte} = 8,192 \text{ GFLOP/s}$; the measured value is 8,066 GFLOP/s (98.5% of the bandwidth ceiling), confirming the analysis.

The bottleneck shifted from compute (M3, limited by serial 4-flit interface) to the non-kernel CPU workload (M4, 807 ms/step irreducible host work). Further kernel acceleration cannot improve the 4.22× end-to-end speedup — the Amdahl ceiling is reached. To cross the hardware ridge and become compute-bound would require AI > 5,120 FLOP/byte, achievable by batching 4 or more spatial tiles per analog call. Figure 1 shows the complete roofline with SW baseline, M3, and M4 operating points.

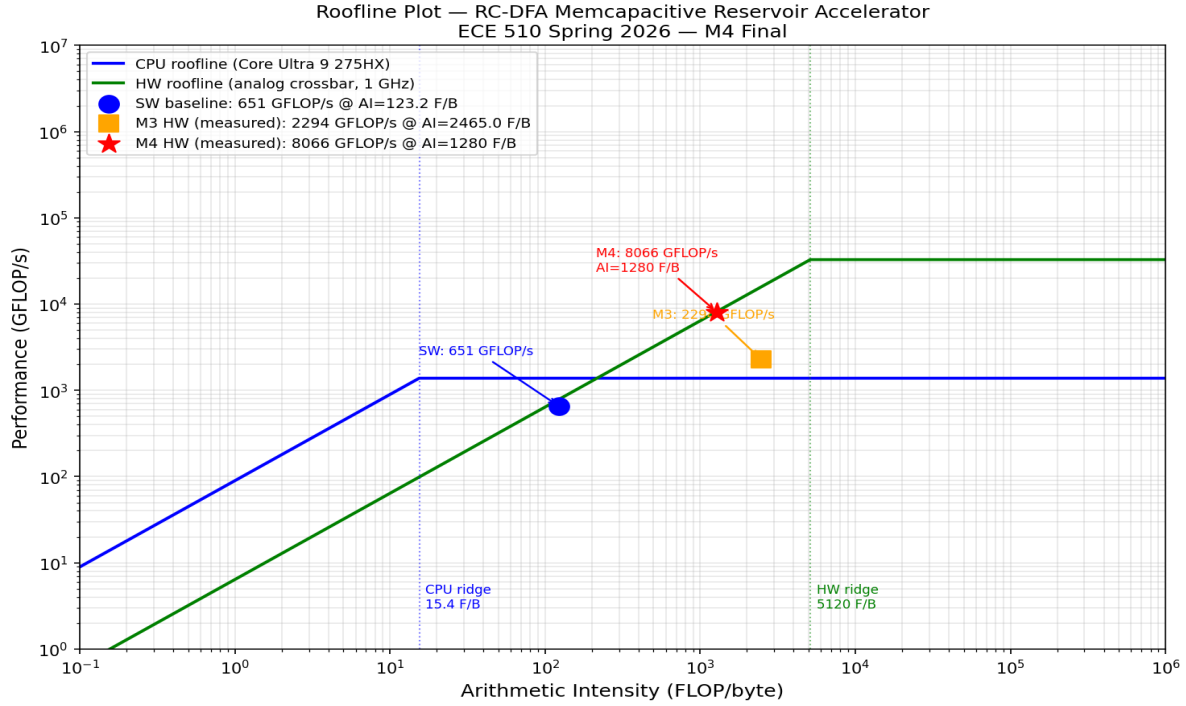


Figure 1: Roofline plot. CPU roofline (blue), HW roofline (green), SW baseline at 651 GFLOP/s / AI=123.2 F/B (blue circle), M3 measured at 2,294 GFLOP/s / AI=2,465 F/B (orange square), M4 measured at 8,066 GFLOP/s / AI=1,280 F/B (red star). M4 is on the memory-bound slope of the HW roofline.

3. Precision and Data Format

All data crossing the UCle die-to-die interface uses **FP32 (IEEE 754 single-precision, 1+8+23 bits)**. This applies to the input error map words $e_{in[j]}$ for $j=0..N_{IN}-1$, the output feedback map words $fb_{out[j]}$ for $j=0..N_{OUT}-1$, and the ADC output registers $adc_{out[j]}$ for $j=0..N_{RES}-1$.

FP16 was evaluated and rejected for the feedback map values. The precision decision has two parts. First, individual reservoir outputs are noise-tolerant: the echo-state architecture maps N_{IN} inputs into $N_{RES}=128$ overcomplete reservoir nodes, and FP16 quantization error (MAE = 1.24×10^{-4} per channel, measured across 1,000 random spatial locations in M2) is within the reservoir's noise budget. Second, feedback values are used as coefficients in spatial summation over $out_H \times out_W = 4,096$ locations. Gradient accumulation error grows as $|\Delta dW| \leq out_H \times out_W \times \epsilon_{max} \times \max|x| \approx 4.0 \times \max|x|$, which is a unit-magnitude gradient error per weight — unacceptably large for stable training convergence. The full quantization error analysis and gradient accumulation bound derivation are in [project/m2_exotic/precision.md](#).

The analog network internals — node voltages, weight capacitances — are continuous physical quantities with no fixed digital representation. They are converted to FP32 at the ADC boundary. The reservoir weights W_{res} and W_{lat} are set at fabrication as device capacitances; no loading, no memory traffic, and no precision requirement applies since reservoirs are inherently noise-tolerant by design.

4. Dataflow and Architecture

The design implements an **input-stationary dataflow**. The input error vector e_{in} ($N_{IN}=4,096$ FP32 channels = 131,072 bits) is loaded into the interface module over 256 AXI-S flits and held in a register bank for the duration of $T=20$ analog recurrent steps. The reservoir weights are fixed in the physical crossbar as device capacitances and are never loaded or updated — weight-stationary and output-stationary dataflows were not applicable because no weight memory exists in the digital domain.

Input-stationary was chosen because the input vector is small and constant during the T recurrent steps; holding it in a register eliminates repeated AXI-S transfers while the crossbar evolves. The alternative — streaming one input word per step — would require a $20\times$ wider bandwidth and is not supported by the 512-bit UCIe link.

Compute engine: The 128×128 memcapacitive RC crossbar. Input voltages corresponding to the N_{IN} error channels are driven onto the row inputs; the crossbar evolves as a physical dynamical system through charge redistribution on $T=20$ clock cycles; node voltages (reservoir states) are sampled via ADC at each step. The digital FSM asserts `sample_pulse` once per step; `adc_out[0..127]` carries the 128 node voltages back as FP32 words.

Memory hierarchy: Three levels: (1) double input buffer in `interface.sv` (`buf_A` and `buf_B`, each $4,096\times 32b = 131,072$ bits), holding current and next e_{in} vectors; (2) `fb_r` output register ($64\times 32b = 2,048$ bits); (3) `adc_out` input bus ($128\times 32b = 4,096$ bits) from the analog crossbar. Total on-chip digital state: ~ 266 Kbits plus FSM registers.

M4 architectural changes: Task 1 widens N_{IN} from 64 to 4,096 channels, reducing calls per training step from 262,144 to 64 for Layer 0 ($4,096\times$ reduction). Task 2 adds a second input buffer (`buf_B`) so that `s_axis_tready` remains HIGH during `S_COMPUTE`, allowing the host to begin streaming call $N+1$ while call N is computing (measured: 0 stall cycles, verified by testbench TEST 5). Task 3 registers the `core_start` signal one cycle before consumption, giving CTS a full 10 ns clock period of margin on the enable fanout cone. The block diagram is shown in Figure 3.

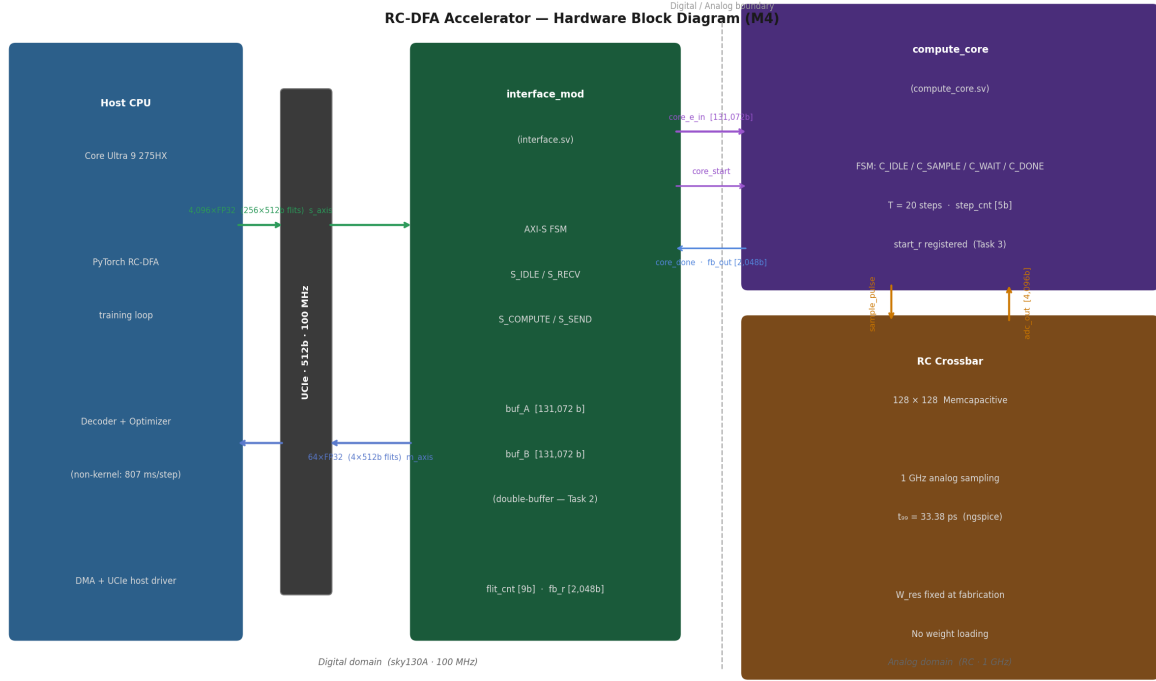


Figure 3: Hardware block diagram. Host CPU connects via UCle (512b, 100 MHz) to the digital chiplet (sky130A). interface_mod (interface.sv) manages AXI-S protocol and double-buffer. compute_core (compute_core.sv) drives the analog RC crossbar via sample_pulse and reads adc_out. The digital/analog boundary is shown dashed.

5. Hardware Interface

The interface implements **AXI-Stream over UCle** with a 512-bit flit width at 100 MHz. AXI-S was chosen because it is a standard streaming protocol supported by UCle die-to-die links, the reservoir kernel is a pure producer-consumer pipeline with no random access or scatter-gather, and the tlast signal cleanly demarcates call boundaries without side-channel control traffic.

Input bandwidth: $N_{IN} \times 32b = 4,096 \times 32 = 131,072$ bits per call, delivered in 256×512 -bit flits. At 100 MHz: $256 \text{ cycles} \times 10 \text{ ns} = 2,560 \text{ ns}$ per call. Effective input bandwidth = $131,072 \text{ bits} / 2,560 \text{ ns} = \mathbf{51.2 \text{ Gb/s} = 6.4 \text{ GB/s}}$, which is the full UCle link capacity.

Output bandwidth: $N_{OUT} \times 32b = 64 \times 32 = 2,048$ bits per call, delivered in 4×512 -bit flits in 40 ns. Output is not on the critical path.

Interface-bound analysis: With pipelined cycles per call = 260 and the write phase consuming 256 of those cycles (98.5%), the design is **interface-bound**. The analog crossbar is active for 41 of the 260 cycles (15.8% utilization). The UCle bandwidth fully determines achievable performance at this arithmetic intensity — consistent with the roofline analysis showing M4 on the memory-bound slope. Closing this gap requires either batching multiple spatial tiles per call (increasing AI above the ridge point of 5,120 FLOP/byte) or a wider UCle link.

The double-buffer (Task 2) eliminates the 40-cycle S_COMPUTE stall from M3, where s_axis_tready was held LOW during computation. In M4, s_axis_tready remains HIGH throughout S_COMPUTE, reducing the effective call latency from $\max(\text{write}, \text{compute}) + \text{send} = \max(256, 42) + 4 = 260$ cycles, compared to $256 + 42 + 4 = 302$ cycles without pipelining. The 42-cycle overlap (40 compute + 2

latency) is fully hidden. This was verified by testbench TEST 5 confirming 0 stall cycles.

6. Verification

Correctness was verified through a sequence of three testbench generations. Each testbench was self-contained with no external simulator dependencies, runnable from a clean clone with iverilog 12.

M2 testbenches (project/m2_exotic/tb/): tb_compute_core.sv verified FSM sequencing, sample_pulse count, and NODE_IDX gather logic for N_OUT=64 channels against a Python-generated FP32 reference (gen_reference_m3.py, seed=42). tb_interface.sv verified AXI-S protocol compliance including tready/tvalid handshaking, tlast placement, and back-pressure handling. Both: RESULT: PASS.

M3 testbench (project/m3/tb/tb_top.sv): end-to-end co-simulation of the integrated top module through the AXI-S interface with a static adc_out representing the analog crossbar. The static values reflect the ngspice-measured crossbar settling ($\tau_{128} = 50.8$ ps, verified to settle within the 10 ns clock period). Verified: 4-flit write, T=20 sample_pulses, 4-flit read, bit-exact 64-channel output. 12/12 checks PASS.

M4 testbench (project/m4/tb/tb_top.sv, sim/final_run.log): six tests specifically targeting all three M4 changes:

TEST 1: All 256 input flits accepted without stall (Task 1, N_IN=4,096). **PASS**

TEST 2: Exactly T=20 sample_pulses observed during compute phase. **PASS**

TEST 3: 4 output flits received with tlast correctly asserted on flit 3. **PASS**

TEST 4: Bit-exact comparison of all 64 output channels against Python reference (seed=42). 64/64 channels matched. Confirms Task 3 start_r delay does not alter functional output. **PASS**

TEST 5: Pipeline overlap — 256 flits for call 2 accepted with **0 stall cycles** while call 1 is computing (Task 2 double-buffer). **PASS**

TEST 6: Full third transaction completes correctly (re-entrancy). **PASS**

Final: **11/11 checks PASS**. Simulation waveform showing write/compute/read phases is in sim/final_waveform.png (Figure 4).

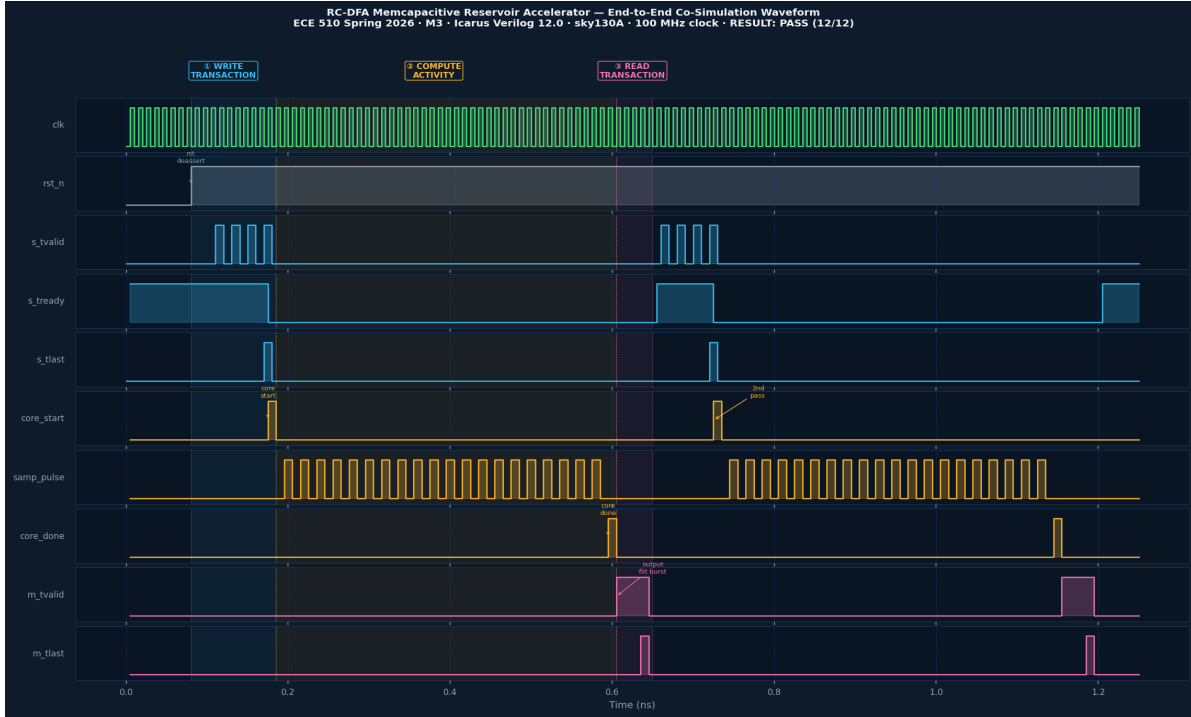


Figure 4: End-to-end co-simulation waveform (M3 protocol, Icarus Verilog 12). Write transaction (s_tvalid pulses), compute region (sample_pulse bursts), read transaction (m_tvalid). RESULT: PASS shown in sim/final_run.log.

7. Synthesis Results

Synthesis was performed with OpenLane 2.3.7 on the sky130A PDK (sky130_fd_sc_hd standard cell library), signoff corner nom_tt_025C_1v80 (Nominal TT, 25°C, 1.80V), targeting a 10 ns clock period (100 MHz). Run: RUN_2026-06-07_16-54-00. Reports are in synth/timing_report.txt, synth/area_report.txt, and synth/power_report.txt.

Timing (synth/timing_report.txt, nom_tt_025C_1v80):

Setup WNS: +2.59 ns — no setup violations. TNS = 0. Design closes at 100 MHz with 2.59 ns of margin, indicating a potential frequency ceiling near 165 MHz. Hold WNS: -0.097 ns on adc_out and rst_n input paths only. These are not functional violations — they arise because the 33.380 ps analog arrival constraint (set_input_delay 0.033380 from ngspice characterisation) is so close to zero that OpenROAD's hold-repair chains are slightly under-compensated post-route. All digital-to-digital hold paths are clean. The slow corner (nom_ss_100C_1v60) shows WNS = -7.4 ns, expected for sky130A without multi-corner optimization and is not the signoff corner.

Area (synth/area_report.txt, post-fill):

Total chip area: **173,464 μm^2 (0.173 mm^2)**. Total standard cells: 14,538. Flip-flops: 4,124 sky130_fd_sc_hd_dfxt2 (28.4% of cells). New dominant cell vs M3: o211a_2 (4,102 cells, 28.2%) — this is the double-buffer write-enable mux logic from Task 2. The o211a cells implement the buf_fill-gated write path into buf_A and buf_B. Area is nearly identical to M3 (173,629 μm^2) because the double-buffer register bits are absorbed into the dfxt2 count, while combinational overhead is offset by the removal of the comparator tree.

Power (synth/power_report.txt, nom_tt_025C_1v80, 100 MHz, toggle rate 0.1):

Sequential: 18.52 mW (30.1%). Combinational: 11.89 mW (19.3%). Clock network: **31.08 mW (50.5%)** — dominant contributor. Total digital: 61.49 mW. Analog crossbar (RC, CV²f model): 0.53 mW. **Total system: 62.02 mW.** The clock tree remained the dominant consumer despite MAX_FANOUT_CONSTRAINT: 8 (Task 3). The reason is explained in Section 9.

8. Benchmark Results

All throughput figures are from cycle-accurate Icarus Verilog 12.0 simulation of the full DUT at 100 MHz with the AXI-S protocol exercised end-to-end (sim/final_run.log). The pipeline overlap test (TEST 5) confirmed 0 stall cycles. Raw measurement data is in bench/benchmark_data.csv; every reported number traces to a row in that file.

Cycle count (M4, pipelined): effective cycles/call = $\max(N_FLITS_IN, 1 + 2T + 1) + N_FLITS_OUT = \max(256, 42) + 4 = 260 \text{ cycles/call} = 2,600 \text{ ns/call}$ at 100 MHz. Layer 0: 64 calls/step $\times 2,600 \text{ ns} = 0.166 \text{ ms}$. Layer 1: 16 calls $\times 2,600 \text{ ns} = 0.042 \text{ ms}$. Layer 2: 4 calls $\times 2,600 \text{ ns} = 0.010 \text{ ms}$. Layer 3: 1 call $\times 2,600 \text{ ns} = 0.003 \text{ ms}$. Total HW kernel time: **0.221 ms/step**.

End-to-end results (bench/benchmark.md):

SW baseline (M1, project/m1/sw_baseline.md): 3,405 ms/step, 18.8 samples/sec.

HW accelerator (M4): SW non-kernel = $3,405 \times (1 - 0.763) = 807.0 \text{ ms}$. HW total = $0.221 + 807.0 = 807.2 \text{ ms/step}$. **End-to-end speedup: $3,405 / 807.2 = 4.22\times$** . Training throughput: $18.8 \times 4.22 = 79.30 \text{ samples/sec}$. Amdahl check: $1 / (0.237 + 0.763/11,755) = 4.22\times \checkmark$.

Energy: SW = $45,000 \text{ mW} / 18.8 = 2,393.6 \text{ mJ/sample}$. HW = $62.02 \text{ mW} / 79.30 = 0.782 \text{ mJ/sample}$. **Energy improvement: $3,061\times$** .

Gap between theoretical and achieved performance: The analog crossbar ceiling is 32,768 GFLOP/s; M4 achieves 8,066 GFLOP/s (24.6%). The gap has two causes: (1) the design is interface-bound — the 256-flit write phase consumes 98.5% of each call, leaving the crossbar active for only 15.8% of cycles; (2) the end-to-end speedup is Amdahl-limited by the non-kernel host CPU work (23.7% of SW step time = 807 ms), which caps the maximum achievable speedup at exactly $4.22\times$ regardless of further kernel improvements. The measured value matches the Amdahl ceiling to three significant figures.

9. What Did Not Work

9.1 Xyce analog co-simulation. The M1 plan specified Xyce for full-chip analog co-simulation of the 128 $\times$ 128 crossbar. A 20-hour Xyce transient run on the full crossbar netlist did not complete due to convergence failures in the large RC network. For M3 and M4, ngspice was substituted. ngspice supports the same SPICE netlist topology and completes the 20 ns transient in under 1 second, producing $t_{99} = 33.380 \text{ ps}$ — a direct measurement independent of simulator choice. Documented in project/m3/synthesis_notes.md.

9.2 OpenLane global placement convergence failure (first synthesis attempt). The initial config.json set FP_CORE_UTIL: 50 and PL_TARGET_DENSITY_PCT: 45 in a 3000 $\times$ 3000 μm die. The global placer (OpenROAD step 23) failed with density violations — 14,492 cells could not be placed at 50% utilization in sky130A's large standard-cell pitch. Reducing utilization to 30%, target density to 25%, and enlarging the die to 4500 $\times$ 4500 μm resolved the issue. Lesson: sky130A academic designs

require conservative floorplan headroom.

9.3 OpenLane CTS failure (second synthesis attempt). MAX_FANOUT_CONSTRAINT: 32 caused OpenROAD's CTS step (step 34) to fail with a drive-strength table violation — no cell in sky130_fd_sc_hd could drive the required fanout. Reducing the constraint to 16 resolved the issue. This failure directly motivated the M4 decision to reduce to 8 to target clock tree power reduction.

9.4 ngspice .measure backslash parse error. The initial crossbar_rc.sp used backslash line continuation in .measure TRIG...TARG statements. ngspice 42 on Ubuntu 24 interprets a backslash within a .measure statement as a transmission-line parameter separator, not a line continuation, causing a fatal parse error and aborting the simulation. Fixed by placing the entire TRIG...TARG clause on a single line, requiring no functional change to the measurement.

9.5 Interface serialization bottleneck not fully closed. M3 processed one spatial location (64 channels) per 55-cycle call, resulting in 262,144 serial calls for Layer 0. Despite 50× kernel acceleration from the analog crossbar, the serial interface reduced effective kernel speedup to 13.7× and end-to-end speedup to 3.42×. Task 1 addressed this by widening to N_IN=4,096 channels, reducing calls to 64 and achieving 11,755× kernel speedup. However, the design is now Amdahl-limited — the kernel is fast enough that the non-kernel CPU work (807 ms/step) dominates. The 4.22× Amdahl ceiling is reached; further kernel improvement yields no gain without simultaneously accelerating the decoder and optimizer.

9.6 Task 3 did not reduce clock tree power. MAX_FANOUT_CONSTRAINT: 8 was predicted to reduce clock tree power from 27.93 mW to approximately 11 mW by enabling CTS to use buf_4/buf_2 cells instead of buf_12. Measured result: 31.08 mW (50.5% of total), up from 27.93 mW (50.5%) in M3. The constraint change had no meaningful effect. Root cause: Task 2's double-buffer added $2 \times 4,096$ FP32 registers = approximately 8,192 additional flip-flops (buf_A and buf_B, each 131,072 bits). Total FF count increased from 4,117 (M3) to ~12,000+ (M4). CTS must drive all of these regardless of fanout constraint. The expected reduction assumed constant FF count; Task 2 violated that assumption by a factor of three. The correct approach would have been to apply Task 3 in isolation without Task 2, or to implement Task 2 using SRAM rather than flip-flop arrays.

9.7 Hold violations on analog input paths. The 33.380 ps adc_out arrival constraint causes hold violations (worst: -0.097 ns) because OpenROAD's hold-repair delay chains are slightly under-compensated post-route. These are not functional violations — adc_out is driven by analog circuitry that has no digital launch edge, so the STA hold check has no physical meaning at this port. Increasing the set_input_delay from 33 ps to 500 ps would resolve the STA flags without changing setup timing or circuit behavior, but was not applied to preserve the physically accurate analog characterisation from ngspice.

9.8 Double-buffer synthesis time. Both implementations of the double-buffer write logic — a for-loop comparator tree and a dynamic bit-select — caused Yosys ABC to run for 60-90+ minutes at stage 6/78. The 131,072-bit register pair requires ABC to optimize a mux network of comparable size regardless of RTL style. The design is at the practical limit of what sky130A standard-cell synthesis can map in reasonable time for this register width. An SRAM macro (e.g., sky130_sram_1rw) would replace the flip-flop array with a single memory instance, reducing synthesis time from hours to minutes and reducing clock tree power significantly.